



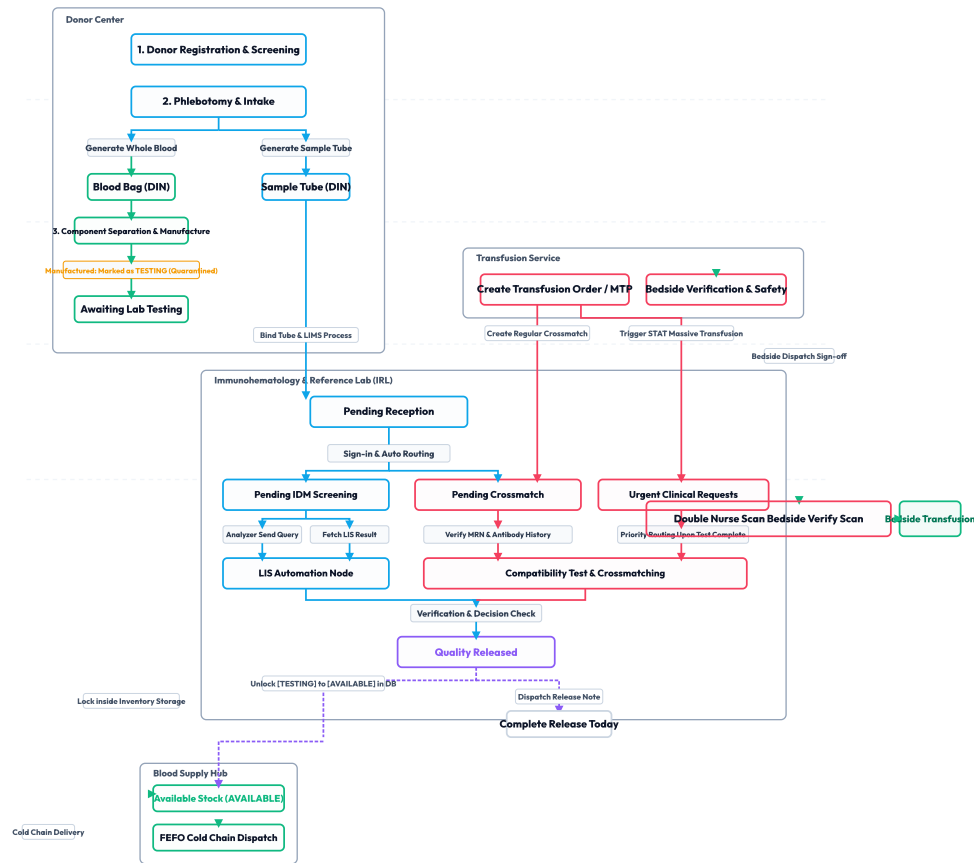
Master System Interoperability Flowchart

VN-BECS V2.0 System Integration Workflows & Safety Logic

Document Type: **Global Interoperability Blueprint**

Review Status: **Approved**

Blue Line - Specimen Flow Green Line - Blood Bag Flow Red Line - Clinical Order Flow



INTEGRATED WORKFLOW

CROSS-SUBSYSTEMS & DATABASE TRANSITIONS

INTEROPERABILITY LOGIC & SAFETY CONTROLS DESCRIPTION

Specimen

Donor Center → IRL
Reception → LIS Auto → QA
Review

1. Specimen Tube Lifecycle & Automated Screening

Sample tubes are linked to DIN at collection and dispatched to IRL. Automation runs NAT/IDM diagnostics, instantly syncing to the BECS database. QA Release releases the specimen and unlocks the blood bags.

INTEGRATED
WORKFLOW

CROSS-SUBSYSTEMS &
DATABASE TRANSITIONS

INTEROPERABILITY LOGIC & SAFETY CONTROLS DESCRIPTION

 Blood Bag

Collection Centrifuge → Hub
TESTING Quarantine → Hub
AVAILABLE → Bedside Scan

2. Blood Product Cold-Chain & Database Lock Controls

Whole blood component separation splits RBC/PLT/Plasma locked as TESTING. IRL QA release unlocks the state to AVAILABLE. Dispatch selects units via FEFO, delivered via IoT cold chain, bedside scan checks.

 Clinical Order

Transfusion Service → IRL
Control Tower → Crossmatch
Verification

3. Transfusion Requisition Matching & STAT Protocols

Orders created in Transfusion Service sync to the IRL control tower queue. STAT MTP triggers flashing red warnings. Technologists perform Electronic (EXM) or Serological (IS-AHG) crossmatching before dispatch.